

Priyanka

Basic Queries:

1. Which origin airports handled the highest number of reported flights during the six-month period?

The screenshot shows a SQL IDE interface. On the left, a 'SCHEMAS' panel lists databases: AirlineDB, airline_project (expanded), music_db, mydb, openflights_db, and sys. Under 'airline_project', there are 'Tables' (staging_fligh..., TimePeriod), 'Views', 'Stored Proced...', and 'Functions'. The main editor displays a SQL query:

```
1 • SELECT
2     ORIGIN,
3     COUNT(*) AS total_departures
4 FROM staging_flights
5 GROUP BY ORIGIN
6 ORDER BY total_departures DESC LIMIT 10;
```

Below the query editor, a 'Result Grid' shows the results of the query. The grid has two columns: 'ORIGIN' and 'total_departur...'. The results are as follows:

ORIGIN	total_departur...
ORD	181265
DEN	156836
ATL	156067
DFW	154180
PHX	102861
LAX	93858
CLT	89572
LAS	85156
MCO	83213
SEA	78980

2. Which flights were cancelled, and what were their date, airline, origin, destination, and diversion status?

The screenshot shows a database management interface with a left sidebar containing a schema tree. The main area displays a SQL query and its results in a table.

SQL Query:

```
1 • SELECT FL_DATE, OP_UNIQUE_CARRIER, ORIGIN, DEST, CANCELLED, DIVERTED
2 FROM staging_flights WHERE CAST(NULLIF(TRIM(CANCELLED), '' ) AS DECIMAL(4,2)) = 1 LIMIT 10;
3
4
```

Result Grid:

FL_DATE	OP_UNIQUE_CARRIER	ORIGIN	DEST	CANCELLED	DIVERTED
1/1/2026 12:00:00 AM	AA	MKE	PHX	1.00	0.00
1/1/2026 12:00:00 AM	AA	DTW	DFW	1.00	0.00
1/1/2026 12:00:00 AM	AA	TUS	DFW	1.00	0.00
1/1/2026 12:00:00 AM	AS	SFO	HNL	1.00	0.00
1/1/2026 12:00:00 AM	AS	HNL	LAS	1.00	0.00
1/1/2026 12:00:00 AM	AS	LAX	PDX	1.00	0.00
1/1/2026 12:00:00 AM	AS	SEA	SJC	1.00	0.00
1/1/2026 12:00:00 AM	AS	LAX	SEA	1.00	0.00
1/1/2026 12:00:00 AM	AS	LAS	SFO	1.00	0.00
1/1/2026 12:00:00 AM	B6	JFK	FLL	1.00	0.00

Advanced Queries:

1. Which origin airports have the highest diversion rates, considering only airports with at least 10,000 flights?

```
WITH airport_statistics AS ( SELECT ORIGIN, COUNT(*) AS total_flights,
SUM( CASE WHEN CAST(NULLIF(TRIM(DIVERTED), '' ) AS DECIMAL(4,2)) = 1
THEN 1
ELSE 0
END ) AS diverted_flights
FROM staging_flights WHERE TRIM(ORIGIN) <> '' GROUP BY ORIGIN
HAVING COUNT(*) >= 10000), ranked_airports AS (SELECT ORIGIN, total_flights,
diverted_flights, ROUND(diverted_flights * 100.0 / total_flights, 2) AS
diversion_rate_percent, RANK() OVER ( ORDER BY diverted_flights * 100.0 /
total_flights DESC) AS airport_rank FROM airport_statistics)
SELECT * FROM ranked_airports WHERE airport_rank <= 10 ORDER BY
airport_rank;
```

SCHEMAS

Filter objects

- AirlineDB
- airline_project
 - Tables
 - staging_fligh...
 - TimePeriod
 - Views
 - Stored Proced...
 - Functions
- music_db
- mydb
- openflights_db
- sys

```

15      HAVING COUNT(*) >= 10000
16    ),
17    ranked_airports AS (
18      SELECT
19        ORIGIN,
20        total_flights,
21        diverted_flights,
22        ROUND(diverted_flights * 100.0 / total_flights, 2)
23          AS diversion_rate_percent,
24        RANK() OVER (
25          ORDER BY diverted_flights * 100.0 / total_flights DESC
26        ) AS airport_rank
27      FROM airport_statistics
28    )
29    SELECT
30      airport_rank,
31      ORIGIN,
32      total_flights,
33      diverted_flights,
34      diversion_rate_percent
35    FROM ranked_airports
36   WHERE airport_rank <= 10
37   ORDER BY airport_rank;
38

```

100% 1:38

Result Grid Filter Rows: Search Export:

airport_ra...	ORIGIN	total_flights	diverted_flights	diversion_rate_percent
1	SNA	22251	102	0.46
2	SAN	49663	204	0.41
3	CHS	12241	50	0.41
4	MIA	58261	224	0.38
5	SJU	18879	72	0.38
6	RSW	20160	73	0.36
7	SAT	19400	64	0.33
8	PHL	49455	158	0.32
9	FLL	48219	154	0.32
10	LAX	93858	299	0.32

2. What were the three busiest origin–destination routes in each month?

```

WITH monthly_routes AS ( SELECT CAST(NULLIF(TRIM(`MONTH`), '') AS
UNSIGNED) AS month_number, ORIGIN, DEST, COUNT(*) AS total_flights
FROM staging_flights WHERE TRIM(ORIGIN) <> '' AND TRIM(DEST) <> ''
GROUP BY CAST(NULLIF(TRIM(`MONTH`), '') AS UNSIGNED), ORIGIN, DEST),
ranked_routes AS ( SELECT month_number,ORIGIN, DEST, total_flights,
ROW_NUMBER() OVER ( PARTITION BY month_number ORDER BY total_flights
DESC) AS route_rank FROM monthly_routes)
SELECT month_number, route_rank, ORIGIN, DEST, total_flights FROM ranked_routes
WHERE route_rank <= 3 ORDER BY month_number, route_rank;

```


Swathi

Basic queries :

1. Compare average air time for A→B and B→A on the same frequently traveled city pair to measure wind/jet-stream impact on fuel and crew planning.

```
12 WITH r AS (  
13     SELECT ORIGIN, DEST, AVG(AIR_TIME) AS avg_air, COUNT(*) AS n  
14     FROM staging_flights  
15     WHERE CANCELLED = 0 AND DIVERTED = 0  
16     GROUP BY ORIGIN, DEST  
17     HAVING n > 2000  
18 )  
19 SELECT a.ORIGIN AS airport_a, a.DEST AS airport_b,  
20        ROUND(a.avg_air) AS a_to_b_min, ROUND(b.avg_air) AS b_to_a_min,  
21        ROUND(a.avg_air - b.avg_air) AS diff_min  
22 FROM r a  
23 JOIN r b ON a.ORIGIN = b.DEST AND a.DEST = b.ORIGIN  
24 WHERE a.ORIGIN < a.DEST  
25 ORDER BY ABS(a.avg_air - b.avg_air) DESC  
26 LIMIT 20;
```

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Result Grid Filter Rows: Search Export:

airport_a	airport_b	a_to_b_min	b_to_a_min	diff_min
BOS	SFO	370	302	68
JFK	SFO	356	296	60
EWR	SFO	342	283	59
BOS	LAX	353	296	57
EWR	LAX	323	276	47
JFK	LAX	334	287	46
DEN	LGA	190	234	-44
ORD	SEA	242	204	39
ATL	LAS	238	200	38
DEN	MCO	182	219	-37
ATL	LAX	259	223	35
ORD	SFO	252	217	35
DFW	SEA	229	198	31
ATL	DEN	173	142	30
BOS	ORD	139	109	30
HNL	LAX	293	323	-30
DFW	MIA	137	164	-28
LAS	ORD	182	209	-27
ORD	PHX	200	175	25
DFW	LGA	177	202	-25

2. Which top 10 scheduled departure hours see the most “late aircraft delay”, so schedulers know where to add buffers.

```

1  SELECT CAST(CRS_DEP_TIME AS UNSIGNED) DIV 100 AS dep_hour,
2         COUNT(*) AS delayed_flights,
3         ROUND(100 * SUM(LATE_AIRCRAFT_DELAY) /
4             SUM(CARRIER_DELAY + WEATHER_DELAY + NAS_DELAY + SECURITY_DELAY + LATE_AIRCRAFT_DELAY), 1) AS pct_late_aircraft
5  FROM staging_flights
6  WHERE LATE_AIRCRAFT_DELAY IS NOT NULL
7  GROUP BY dep_hour
8  ORDER BY delayed_flights desc
9  LIMIT 10;

```

	dep_hour	delayed_flights	pct_late_aircraft
17	63111	43.9	
18	61802	47.0	
19	58362	48.5	
16	57045	43.6	
15	52450	42.9	
14	49541	41.8	
20	48155	49.1	
13	47119	42.3	
12	45237	38.9	
11	41166	38.6	

Advanced queries :

1. Track how the cancellation rate changes from one month to the next, and how much of it is weather-driven, to separate seasonal shocks from operational problems.

```

54  WITH monthly AS (
55      SELECT MONTH,
56             COUNT(*) AS flights,
57             ROUND(100 * AVG(CANCELLED), 2) AS cancel_pct,
58             ROUND(100 * SUM(CANCELLATION_CODE = 'B') / NULLIF(SUM(CANCELLED), 0), 1) AS weather_share_pct
59      FROM staging_flights
60      GROUP BY MONTH
61  )
62  SELECT m.MONTH, m.flights, m.cancel_pct, m.weather_share_pct,
63         ROUND(m.cancel_pct - (SELECT p.cancel_pct FROM monthly p WHERE p.MONTH = m.MONTH - 1), 2) AS mom_change_pts
64  FROM monthly m
65  ORDER BY m.MONTH;
66

```

	MONTH	flights	cancel_pct	weather_share_pct	mom_change_pts
1	544003	4.71	87.6	NULL	
2	515037	2.25	85.4		-2.46
3	612102	2.90	56.4		0.65
4	597919	0.90	56.1		-2.00
5	611735	0.92	58.7		0.02
6	607577	1.65	65.9		0.73

2. Which routes let a 30+ minute late departure still arrive on time, more often than the network average? That would indicate padded block times.

```
107 • SELECT ORIGIN, DEST, COUNT(*) AS late_deps,  
108        ROUND(100 * AVG(ARR_DEL15 = 0), 1) AS recovery_pct  
109 FROM staging_flights  
110 WHERE DEP_DELAY >= 30 AND CANCELLED = 0 AND DIVERTED = 0  
111 GROUP BY ORIGIN, DEST  
112 HAVING late_deps >= 50  
113        AND recovery_pct > (SELECT 100 * AVG(ARR_DEL15 = 0)  
114                             FROM staging_flights  
115                             WHERE DEP_DELAY >= 30 AND CANCELLED = 0 AND DIVERTED = 0)  
116 ORDER BY recovery_pct DESC  
117 LIMIT 5;  
118
```

100% 1:108

Result Grid



Filter Rows:



Search

Export:



Fetch rows:



ORIGIN	DEST	late_deps	recovery_pct
IAD	PHX	68	29.4
KOA	SEA	50	26.0
BWI	LAS	146	25.3
BDL	DFW	62	24.2
SEA	EWR	118	23.7

Charishma

Basic queries:

1. List the top 10 airlines that have the best on-time arrival performance.

```
SELECT
    OP_UNIQUE_CARRIER AS Airline,
    ROUND(AVG(CAST(ARR_DELAY AS DECIMAL(10,2))), 2) AS AvgArrivalDelay
FROM staging_flights
WHERE
    ARR_DELAY IS NOT NULL
    AND ARR_DELAY <> ''
    AND CANCELLED = '0.00'
GROUP BY OP_UNIQUE_CARRIER
ORDER BY AvgArrivalDelay ASC
LIMIT 10;
```

Airline	AvgArrivalDelay	
AS	1.68	
DL	2.71	
YX	3.10	
MQ	3.82	
UA	4.76	
WN	6.71	
OO	10.23	
B6	10.84	
AA	11.99	
F9	12.28	

2. List the top 10 origin airports that experience the most departure delays.

```
SELECT
    ORIGIN as AirportCode,
    ORIGIN_CITY_NAME AS City,
    COUNT(*) AS TotalFlights,
    ROUND(AVG(CAST(DEP_DELAY AS DECIMAL(10,2))), 2) AS AvgDepartureDelay
FROM staging_flights
WHERE
    DEP_DELAY IS NOT NULL
    AND DEP_DELAY <> ''
    AND CANCELLED = '0.00'
GROUP BY
    ORIGIN,
    ORIGIN_CITY_NAME
HAVING COUNT(*) > 1000
ORDER BY AvgDepartureDelay DESC
LIMIT 10;
```

AirportCode	City	TotalFlights	AvgDepartureDelay	
ASE	Aspen, CO	4146	27.36	
FWA	Fort Wayne, IN	2877	25.33	
SBN	South Bend, IN	3902	25.10	
FNT	Flint, MI	1797	24.97	
XWA	Williston, ND	1115	24.76	
CRW	Charleston/Dunbar, WV	1223	24.46	
LFT	Lafayette, LA	1128	24.36	
EGE	Eagle, CO	2630	23.99	
CAK	Akron, OH	2453	22.48	
HDN	Hayden, CO	1522	21.87	

Advanced queries:

1. Which airlines perform worse than the overall average arrival delay?

```
SELECT
    OP_UNIQUE_CARRIER AS Airline,
    ROUND(AVG(CAST(ARR_DELAY AS DECIMAL(10,2))), 2) AS AvgDelay
FROM staging_flights
WHERE
    ARR_DELAY <> ''
    AND CANCELLED = '0.00'
GROUP BY OP_UNIQUE_CARRIER
HAVING AVG(CAST(ARR_DELAY AS DECIMAL(10,2)))
>
(
    SELECT AVG(CAST(ARR_DELAY AS DECIMAL(10,2)))
    FROM staging_flights
    WHERE
        ARR_DELAY <> ''
        AND CANCELLED = '0.00'
)
ORDER BY AvgDelay DESC;
```

	Airline	AvgDelay	
	NK	19.46	
	G4	15.96	
	OH	14.74	
	F9	12.28	
	AA	11.99	
	B6	10.84	
	OO	10.23	

2. Which origin airports are served by the greatest number of different airlines, and how many flights and destinations did they have?

```
SELECT
    ORIGIN AS Airport,
    COUNT(DISTINCT OP_UNIQUE_CARRIER) AS NumberOfAirlines,
    COUNT(*) AS TotalFlights,
    COUNT(DISTINCT DEST) AS NumberOfDestinations
FROM staging_flights
WHERE CANCELLED = '0.00'
GROUP BY ORIGIN
HAVING COUNT(DISTINCT OP_UNIQUE_CARRIER) = (
    SELECT MAX(AirlineCount)
    FROM (
        SELECT
            ORIGIN,
            COUNT(DISTINCT OP_UNIQUE_CARRIER) AS AirlineCount
        FROM staging_flights
        WHERE CANCELLED = '0.00'
        GROUP BY ORIGIN
    ) AS AirportAirlines
);
```

	Airport	NumberOfAirlines	TotalFlights	NumberOfDestinations	
	AUS	13	43803	81	
	BNA	13	53066	102	
	CHS	13	11820	28	
	MSY	13	24362	41	
	PIT	13	20123	47	